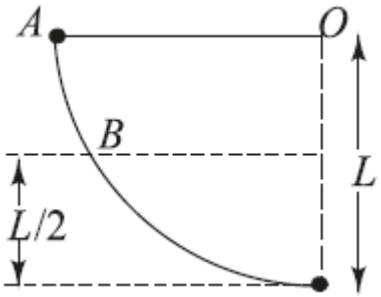
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A steel ball of mass m tied to a light string of length L is released from rest when the string is horizontal. The tension in the string when the ball is at a height $\frac{L}{2}$ above the lowest position (see fig.) is



(A) $\frac{1}{2}mg$

(B) $\sqrt{2}mg$

(C) $\frac{3}{2}mg$

(D) $2mg$

Q.2

A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time ' t ' is proportional to

(A) $t^{3/4}$

(B) $t^{3/2}$

(C) $t^{1/4}$

(D) $t^{1/2}$

Q.3

A particle of mass m is driven by a machine that delivers a constant power of k watts. If the particle starts from rest the force on the particle at time t is

(A) $\sqrt{\mathrm{mk}}\mathrm{t}^{-1/2}$

(B) $\sqrt{2\mathrm{mk}}\mathrm{t}^{-1/2}$

(C) $\frac{1}{2}\sqrt{\mathrm{mk}}\mathrm{t}^{-1/2}$

(D) $\sqrt{\frac{\mathrm{mk}}{2}}\mathrm{t}^{-1/2}$

Q.4

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A body of mass $1\ \mathrm{kg}$ begins to move under the action of a time dependent force $\overrightarrow{\mathrm{F}}=\left(2t\hat{\mathrm{i}}+3\mathrm{t}^2\hat{\mathrm{j}}\right)\ \mathrm{N}$ where i and $\hat{\mathrm{j}}$ are unit vectors along x and y axis. What power will be developed by the force at the time t?

(A) $\left(2t^2+3t^3\right)\ W$

(B) $\left(2t^2+4t^4\right)\ W$

(C) $\left(2t^3+3t^4\right)\ W$

(D) $\left(2t^3+3t^5\right)\ W$

Q.5

A body of mass $1\ \mathrm{kg}$ begins to move under the action of a time dependent force $\overrightarrow{\mathrm{F}}=\left(2t\hat{\mathrm{i}}+3\mathrm{t}^2\hat{\mathrm{j}}\right)\ \mathrm{N}$, where $\hat{\mathrm{i}}$ and $\hat{\mathrm{j}}$ are
unit vectors alogn x and yaxis. What power will be developed by the force at the time t ?

(A) $\left(2t^2+3t^3\right)\ W$

(B) $\left(2t^2+4t^4\right)\ W$

(C) $\left(2t^3+3t^4\right)\ W$

(D) $\left(2t^3+3t^5\right)\ W$

Q.6

A car of mass m starts from rest and accelerates so that the instantaneous power delivered to the car has a constant magnitude P_0. The instantaneous velocity of this car is proportional to:

(A) $\mathrm{t}^2\mathrm{P}_0$

(B) $\mathrm{t}^{1/2}$

(C) $t^{-1/2}$

(D) $\frac{t}{\sqrt{m}}$

Q.7

A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time ' t ' is proportional to

(A) $t^{3/4}$

(B) $t^{3/2}$

(C) $t^{1/4}$

(D) $t^{1/2}$

Q.8

A body projected vertically from the earth reaches a height equal to earth's radius before returning to the earth. The

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power exerted by the gravitational force is greatest

(A) at the highest position of the body

(B) at the instant just before the body hits the earth

(C) it remains constant all through

(D) at the instant just after the body is projected

Q.9

A constant power P is applied to a car starting from rest. If v is the velocity of the car at time t, then

(A) $v \propto t$

(B) $v \propto \frac{1}{t}$

(C) $v \propto \sqrt{t}$

(D) $v \propto \frac{1}{\sqrt{t}}$

Q.10

If two persons A and B take 2 seconds and 4 seconds respectively to lift an object to the same height h, then the ratio of their powers is

(A) $1:2$

(B) $1:1$

(C) $2:1$

(D) $1:3$

Q.11

A car of weight W is on an inclined road that rises by $100\ \mathrm{m}$ over a distance of $1\mathrm{Km}$ and applies a constant frictional force $\frac{\mathrm{W}}{20}$ on the car. While moving uphill on the road at a speed of $10\ \mathrm{ms}^{-1}$, the car needs power P. If it needs power $\frac{\mathrm{P}}{2}$ while moving downhill at speed v then value of v is:

(A) $20\ \mathrm{ms}^{-1}$

(B) $5\ \mathrm{ms}^{-1}$

(C) $15\ \mathrm{ms}^{-1}$

(D) $10\ \mathrm{ms}^{-1}$

Q.12

If a machine gun fires n bullets per second each with kinetic energy K, then the power of the machine gun is

(A) $n\ \mathrm{K}^2$

(B) $\frac{\mathrm{K}}{\mathrm{n}}$

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(C) $\mathrm{n^2\ K}$

(D) nK

Q.13

An engineer claims to have made an engine delivering $10\ \mathrm{kW}$ power with fuel consumption of $1\ \mathrm{g/s}$. The calorific value of fuel is $2\mathrm{kcal/g}$. This claim is

(A) valid

(B) invalid

(C) depends on engine design

(D) dependent on load

Q.14

Water falls from a height of $60\ \mathrm{m}$ at the rate of $15\ \mathrm{kg/s}$ to operate a turbine. The losses due to frictional force are 10% of energy. How much power is generated by the turbine? $(\mathrm{g} = 10\ \mathrm{m/s^2})$

(A) $8.1\ \mathrm{kW}$

(B) $10.2\ \mathrm{kW}$

(C) $12.3\ \mathrm{kW}$

(D) $7.0\ \mathrm{kW}$

Q.15

A $10\ \mathrm{m}$ long iron chain of linear mass density $0.8\ \mathrm{kgm^{-1}}$ is hanging freely from a rigid support. If $\mathrm{g} = 10\ \mathrm{ms^{-2}}$, then the power required to left the chain upto the point of support in 10 second

(A) $10\ \mathrm{W}$

(B) $20\ \mathrm{W}$

(C) $30\ \mathrm{W}$

(D) $40\ \mathrm{W}$

Q.16

A wind-powered generator converts wind energy into electrical energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v, the electrical power output will be proportional to

(A) v

(B) v^2

(C) v^3

(D) v^4

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Q.17

A particle of mass $10\ \mathrm{g}$ moves along a circle of radius $6.4\ \mathrm{cm}$ with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to $8 \times 10^{-4}\ \mathrm{J}$ by the end of the second revolution after the beginning of the motion ?

(A) $0.1\ \mathrm{m/s^2}$

(B) $0.15\ \mathrm{m/s^2}$

(C) $0.18\ \mathrm{m/s^2}$

(D) $0.2\ \mathrm{m/s^2}$

Q.18

A stone is tied to a string of length ℓ and is whirled in a vertical circle with the other end of the string as the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of the change in velocity as it reaches a position where the string is horizontal (g being acceleration due to gravity) is

(A) $\sqrt{2\ \mathrm{g}\ell}$

(B) $\sqrt{2\left(\mathrm{u}^2 - \mathrm{g}\ell\right)}$

(C) $\sqrt{\mathrm{u}^2 - \mathrm{g}\ell}$

(D) $\mathrm{u} - \sqrt{\mathrm{u}^2 - 2\ \mathrm{g}\ell}$

Q.19

A body of mass $10\ \mathrm{kg}$ moves with a velocity v of $2\ \mathrm{m/s}$ along a circular path of radius $8\ \mathrm{m}$. The power produced by the body will be

(A) $10\ \mathrm{J/s}$

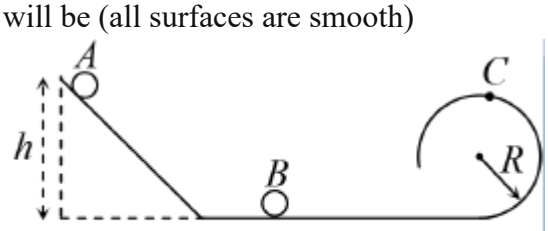
(B) $98\ \mathrm{J/s}$

(C) $49\ \mathrm{J/s}$

(D) zero

Q.20

Ball A of mass m after sliding from an inclined plan strikes elastically with another ball B of mass $3\ \mathrm{m}$ at rest. The minimum height h, so that the ball B just completes the circular portion of the surface at C, will be (all surfaces are smooth)



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(A) $h = \frac{5}{2}R$

(B) $h = 5R$

(C) $h = 8R$

(D) $h = 10R$

Answers & Solutions

Q.1 Answer:

$\frac{3}{2}mg$

Solution:

Refer to Fig.

The equation of motion when the ball is at position B is

$\frac{mv^2}{L} = T - mg\cos\theta$ **(1)**

Where v is the speed of the ball at point B. From conservation of energy,

Total energy at A = total energy at B

$mgL + 0 = \frac{1}{2}mv^2 + mg\left(\frac{L}{2}\right)$

$\Rightarrow \quad mv^2 = mgL$ **(2)**

Using (2) in (1), we have

$mg = T - mg\cos\theta$

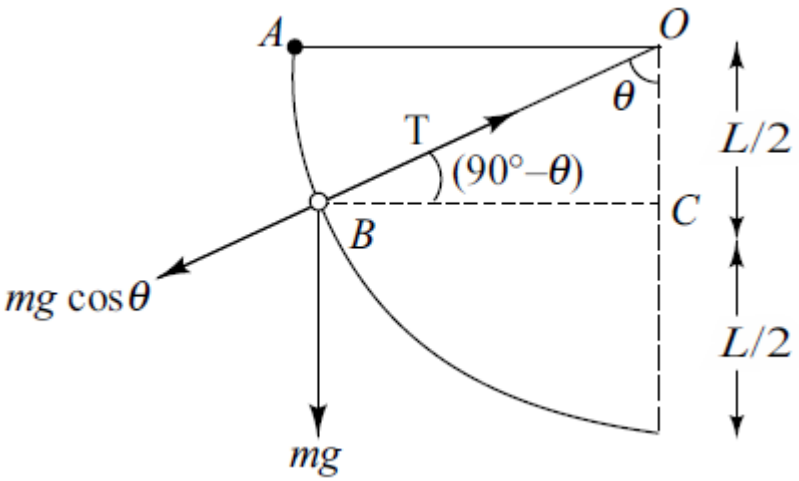
$\Rightarrow \quad T = mg(1 + \cos\theta)$

In triangle OBC $\sin(90^\circ - \theta) = \frac{OC}{OB} = \frac{L/2}{L} = \frac{1}{2}$

$\Rightarrow \quad \cos\theta = \frac{1}{2}$

$\therefore \quad T = mg\left(1 + \frac{1}{2}\right) = \frac{3}{2}mg$

So the correct choice is (c).



Q.2 Answer:

$t^{3/2}$

Solution:

(b) We know that $\mathrm{F} \times \mathrm{v}$ **= Power** $\therefore F \times v = c$ **where** c **= constant** $\therefore m\frac{dv}{dt} \times v = c \quad \left(\therefore F = ma = \frac{mdv}{dt}\right)$

$\therefore m\int_0^v vdv = c\int_0^t dt \quad \therefore \frac{1}{2}mv^2 = ct$

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$\therefore v = \sqrt{\frac{2c}{m}} \times t^{1/2}$

$\therefore \frac{dx}{dt} = \sqrt{\frac{2c}{m}} \times t^{1/2}$ **where** $v = \frac{dx}{dt}$

$\therefore \int_0^x dx = \sqrt{\frac{2c}{m}} \times \int_0^t t^{1/2} dt$

$x = \sqrt{\frac{2c}{m}} \times \frac{2t^{3/2}}{3} \Rightarrow x \propto \mathrm{t}^{3/2}$

Q.3 Answer:

$\sqrt{\frac{\mathrm{mk}}{2}}\mathrm{t}^{-1/2}$

Solution:

As we know power $\mathrm{P} = \frac{\mathrm{dw}}{\mathrm{dt}}$

$$\Rightarrow \quad \mathrm{w} = \mathrm{Pt} = \frac{1}{2}\mathrm{mv}^2$$

So, $v = \sqrt{\frac{2Pt}{m}}$

Hence, acceleration $\mathrm{a} = \frac{\mathrm{dv}}{\mathrm{dt}} = \sqrt{\frac{2\mathrm{P}}{\mathrm{m}}} \cdot \frac{1}{2\sqrt{\mathrm{t}}}$

Therefore, force on the particle at time ' t **'**

$$= ma = \sqrt{\frac{2\mathrm{Km}^2}{\mathrm{m}}} \cdot \frac{1}{2\sqrt{\mathrm{t}}} = \sqrt{\frac{\mathrm{Km}}{2\mathrm{t}}} = \sqrt{\frac{\mathrm{mK}}{2}}\mathrm{t}^{-1/2}$$

Q.4 Answer:

$\left(2t^3 + 3t^5\right) W$

Solution:

Given force $\vec{F} = 2t\hat{i} + 3t^2\hat{j}$

According to Newton's second law of motion, $m\frac{d\vec{v}}{dt} = 2t\hat{i} + 3t^2\hat{j} \quad (\mathrm{m} = 1\ \mathrm{kg})$

$\Rightarrow \quad \int_0^{\vec{v}} d\vec{v} = \int_0^t \left(2t\hat{i} + 3t^2\hat{j}\right) dt \Rightarrow \vec{v} = t^2\hat{i} + t^3\hat{j}$

Power $\mathrm{P} = \vec{F} \cdot \vec{v} = \left(2t\hat{i} + 3t^2\hat{j}\right) \cdot \left(t^2\hat{i} + t^3\hat{j}\right)$

$= \left(2t^3 + 3t^5\right) W$

Q.5 Answer:

$\left(2t^3 + 3t^5\right) W$

Solution:

Given force $\overrightarrow{\mathrm{F}} = 2\mathrm{t}\hat{\mathrm{i}} + 3\mathrm{t}^2\hat{\mathrm{j}}$

According to Newton's second law of motion, $\mathrm{m}\frac{\mathrm{d}\vec{\mathrm{v}}}{\mathrm{dt}} = 2\hat{\mathrm{t}} + 3\mathrm{t}^2\hat{\mathrm{j}} \quad (\mathrm{m} = 1\ \mathrm{kg})$

$\Rightarrow \quad \int_0^{\vec{v}} d\vec{v} = \int_0^t \left(2t\hat{i} + 3t^2\hat{j}\right) dt$

$\Rightarrow \quad \overrightarrow{\mathrm{v}} = \mathrm{t}^2\hat{\mathrm{i}} + \mathrm{t}^3\hat{\mathrm{j}}$

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Power $\mathrm{P}=\overrightarrow{\mathrm{F}} \cdot \overrightarrow{\mathrm{v}}=\left(2 \mathrm{t} \hat{\mathrm{i}}+3 \mathrm{t}^{2} \hat{\mathrm{j}}\right) \cdot\left(\mathrm{t}^{2} \hat{\mathrm{i}}+\mathrm{t}^{3} \hat{\mathrm{j}}\right)$
$=\left(2 \mathrm{t}^{3}+3 \mathrm{t}^{5}\right) \mathrm{W}$

Q.6 Answer:

$\mathrm{t}^{1/2}$

Solution:

Constant power of car $P_0 = F.v = \mathrm{ma.v}$

$$P_0 = m\frac{dv}{dt}.v$$
$$P_0 dt = mvdv \text{ Integrating}$$
$$P_0 \cdot t = \frac{mv^2}{2}$$
$$v = \sqrt{\frac{2P_0 t}{m}}$$

$\because$ P_0, m **and 2 are constant**

$$\therefore \quad v \propto \sqrt{t}$$

Q.7 Answer:

$t^{3/2}$

Solution:

We know that $\mathbf{F} \times \mathrm{v}=$ **Power** $\therefore F \times v = c$ **where** $\mathrm{c}=$ **constant** $m\frac{dv}{dt} \times v =$
$c \quad \left(\because F = ma = \frac{mdv}{dt}\right)$
$m\int_0^v vdv = c\int_0^t dt \Rightarrow \frac{1}{2}mv^2 = ct$
$v = \sqrt{\frac{2c}{m}} \times t^{1/2}$
$\frac{dx}{dt} = \sqrt{\frac{2c}{m}} \times t^{1/2}$ **where** $v = \frac{dx}{dt}$
$\int_0^x dx = \sqrt{\frac{2c}{m}} \times \int_0^t t^{1/2}dt$
$x = \sqrt{\frac{2c}{m}} \times \frac{2t^{3/2}}{3} \Rightarrow \mathrm{x} \propto \mathrm{t}^{3/2}$

Q.8 Answer:

at the instant just before the body hits the earth

Solution:

Power exerted by a force is given by $\mathrm{P}=\mathrm{F} \cdot \mathrm{V}$

When the body is just above the earth's surface, its velocity is greatest. At this instant, gravitational force is also maximum. Hence, the power exerted by the gravitational force is greatest at the instant just before the body hits the earth.

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Q.9 Answer:

$v \propto \sqrt{t}$

Solution:

$$\text{Power} = \frac{\text{work done}}{\text{time}} = \frac{\frac{1}{2}mv^2}{t}$$

$$p \times t = \frac{1}{2}mv^2$$

$$\therefore v^2 \propto t$$

$$\Rightarrow v \propto \sqrt{t}$$

Q.10 Answer:

$2 : 1$

Solution:

Power $= \frac{\text{work done}}{\text{time}}$

Therefore power of $\mathrm{A}, \mathrm{P_A} = \frac{\mathrm{mgh}}{\mathrm{t_A}}$

and power of $\mathrm{B}, \mathrm{P_B} = \frac{\mathrm{mgh}}{\mathrm{t_B}}$

$\therefore \quad \frac{\mathrm{P_A}}{\mathrm{P_B}} = \frac{\mathrm{t_B}}{\mathrm{t_A}} = \frac{4}{2} = 2 : 1$

Q.11 Answer:

$15\ \mathrm{ms}^{-1}$

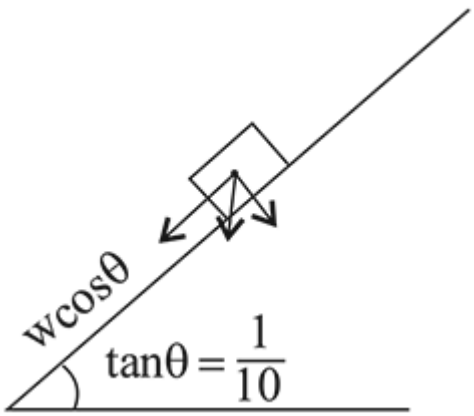
Solution:

While moving downhill power $P = \left(w \sin\theta + \frac{w}{20}\right) 10$

$P = \left(\frac{w}{10} + \frac{w}{20}\right) 10 = \frac{3w}{2}$

$\frac{P}{2} = \frac{3w}{4} = \left(\frac{w}{10} - \frac{w}{20}\right) V$

$\frac{3}{4} = \frac{v}{20} \Rightarrow v = 15\ \mathrm{m/s} \quad \$100\%\ \tan\theta = \frac{1}{10}$ \ **Speed of car while moving downhill** $\mathrm{v} = 15\ \mathrm{m/s}$



Q.12 Answer:

nK

Solution:

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$$\begin{aligned}\text{Power } &= \frac{\text{total work done}}{\text{time}} \\ &= \frac{\frac{1}{2}\mathrm{Mv}^2}{\mathrm{t}} = \frac{1}{2}\left(\mathrm{mv}^2\right)\mathrm{n}\left(\because \frac{\mathrm{M}}{\mathrm{t}} = \mathrm{mn}\right) \\ &= \mathrm{kn}\left[\because \mathrm{K.E.K} = \frac{1}{2}\mathrm{mv}^2\right.\end{aligned}$$

Q.13 Answer:

invalid

Solution:

$\mathrm{P} = 10\ \mathrm{kW} = 10000\ \mathrm{W}$

Fuel consumption $= 1\ \mathrm{g/s}$ **Calorific value** $= 2\mathrm{kcal/g}$

$\therefore$ **Energy produced** $= 2\mathrm{kcal/s}$

Input power $= 2\mathrm{kcal/s} = 2000\mathrm{cal/s}$

$= 2000 \times 4.18\ \mathrm{J/s} = 8.4\mathrm{kW}$

This claim is invalid.

Q.14 Answer:

$8.1\ \mathrm{kW}$

Solution:

Given, $\mathrm{h} = 60\ \mathrm{m},\ \mathrm{g} = 10\ \mathrm{ms}^{-2}$,

Rate of flow of water $= 15\ \mathrm{kg/s}$

$\therefore$ **Power of the falling water** $= 15\mathrm{kgs}^{-1} \times 10\ \mathrm{ms}^{-2} \times 60\ \mathrm{m} = 900$ **watt.**

Loss in energy due to friction

$= 9000 \times \frac{10}{100} = 900\mathrm{watt.}$

$\therefore$ **Power generated by the turbine** $= (9000 - 900)$ **watt** $= 8100$ **watt** $= 8.1\ \mathrm{kW}$

Q.15 Answer:

$40\ \mathrm{W}$

Solution:

$\mathrm{m} = 10 \times 0.8\ \mathrm{kg} = 8\ \mathrm{kg}$

height of iron chain $= 5\ \mathrm{m}\ \mathrm{P} = \frac{\mathrm{mgh}}{\mathrm{t}} = \frac{8 \times 10 \times 5}{10}\ \mathrm{W} = 40\ \mathrm{W}$

Q.16 Answer:

v^3

Solution:

Number of particles string the blades/time

$\propto$ **velocity of wind**

K.E. of particle $\propto$ **(Velocity of wind)**

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$\therefore$ **Power output** $\propto$ **(No. of particles striking/ time** $) \times (\mathrm{K.E.}$ **of particle** $)$

$\therefore$ **Power output** $\propto v^3$

Q.17 Answer:

$0.1\ \mathrm{m/s^2}$

Solution:

Given: Mass of particle, $\mathrm{M} = 10\ \mathrm{g} = \frac{10}{1000}\ \mathrm{kg}$

radius of circle $\mathrm{R} = 6.4\ \mathrm{cm}$ **Kinetic energy E of particle** $= 8 \times 10^{-4}\ \mathrm{J}$ **acceleration** $\mathrm{a_t} = ?$ $\frac{1}{2}mv^2 = E$

$\Rightarrow \quad \frac{1}{2}\left(\frac{10}{1000}\right)\mathrm{v}^2 = 8 \times 10^{-4}$

$\Rightarrow \quad v^2 = 16 \times 10^{-2}$

$\Rightarrow \quad v = 4 \times 10^{-1} = 0.4\ \mathrm{m/s}$

Now, using $\mathrm{v}^2 = \mathrm{u}^2 + 2\mathrm{a_t s} \quad (\mathrm{s} = 4\pi\mathrm{R})$

$(0.4)^2 = 0^2 + 2\mathrm{a_t}\left(4 \times \frac{22}{7} \times \frac{6.4}{100}\right)$

$\Rightarrow \quad a_\mathrm{t} = (0.4)^2 \times \frac{7\times100}{8\times22\times6.4} = 0.1\ \mathrm{m/s^2}$

Q.18 Answer:

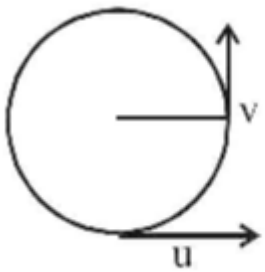
$\sqrt{2\left(\mathrm{u}^2 - \mathrm{g}\ell\right)}$

Solution:

$\mathrm{W_{mg}} = \Delta\mathrm{K} \Rightarrow -\mathrm{mg}\ell = 1/2\mathrm{mv}^2 - 1/2\mathrm{mu}^2$

or $\mathrm{mv}^2 = \mathrm{m}\left(\mathrm{u}^2 - 2\ \mathrm{g}\ell\right)$ **or** $\mathrm{v} = \sqrt{\mathrm{u}^2 - 2\ \mathrm{g}\ell}\hat{\mathrm{j}}$

$\overrightarrow{\mathrm{u}} = \mathrm{u}\hat{\mathrm{i}}$



$\therefore \vec{v} - \vec{u} = \sqrt{u^2 - 2g\ell}\hat{j} - u\hat{i}$

$\therefore |\vec{v} - \vec{u}| = \left[\left(u^2 - 2g\ell\right) + u^2\right]^{1/2} = \sqrt{2\left(u^2 - g\ell\right)}\Big)$

Q.19 Answer:

zero

Solution:

The power of body is given by $= \vec{F} \cdot \vec{v}$ **as**

the body is moving in circular path, centripetal force and velocity are at 90°**, or power** $= 0$

Q.20 Answer:

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$h = 10R$

Solution:

$u_A = \sqrt{2gh}, \quad v_B = \frac{2m_1 u_A}{m_1+m_2} = \frac{u_A}{2} = \frac{\sqrt{2gh}}{2}$

To complete vertical circle, $v_B^2 = 5gR \quad \Rightarrow \quad \frac{2gh}{4} = 5gR$

$\therefore \quad h = 10R$

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